



First Semester 2026-27

Course Handout Part II

Date: 03082026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : CHEM F212

Course Title : Organic Chemistry I

Instructor-in-charge : PARITOSH SHUKLA

Course description: Basic terminology and representation of organic reactions; thermodynamics and kinetics of reactions; reactive intermediates (carbocations, carbanions, free radicals, nitrenes carbenes); aromatic chemistry; properties, preparation and reactions of alkyl halides, alcohols, ethers, amines and nitro compounds; carbonyl compounds; carboxylic acid and derivatives; carbohydrates

1. Scope and Objective: Basic organic chemistry and functional group chemistry. Course aims to develop critical thinking for organic transformations and logical understanding of synthetic methods in organic chemistry.

2. Course learning outcomes: After completing this course student will

- understand the concepts of electron pushing/pulling to the fundamental organic reaction.
- understand the methodologies that lead to proposing a reaction mechanism.
- understand functional group chemistry, devise organic reactions, and determine organic products formed *via* substitution, elimination, addition and rearrangement reactions and associated mechanisms.
- employ critical thinking and efficient problem-solving skills in organic chemistry.

3. Text books and references:

TB. R. T. Morrison, R. Boyd and S. K. Bhattacharjee, Organic Chemistry, 7th edition.

RB1: W J. Clayden, N. Greeves, S. Warren, P. Wothers, Organic Chemistry, 2nd Edition

RB2: G. Marc Loudon, Organic Chemistry, 4th edition

4. Course Plan:

Module	Topics to be covered	Reference	Learning Outcomes
1. Basic terminology, representation of organic reactions and methods for determining reaction mechanism.	L1. Homolytic, heterolytic fission of bonds, concept of electrophiles and nucleophiles, curved and fish-hook arrows, movement of arrows L2. Methods for determining mechanisms: identification of reaction	T1: Ch. 4, pg. 55-59. R1: Ch. 5, pg. 116-123. Lecture notes	Write the mechanism of simple organic reactions.



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	intermediate, stereochemical evidence etc. L3. Isotope labeling & kinetic isotopic effect. T1. Exercise problems on determination and drawing of simple reaction mechanism.		
2. Kinetics and thermodynamic aspects of organic reactions	L4-5. Importance of thermodynamic and kinetics in controlling the reaction pathway L6. Hammond postulate & its application in understanding reaction mechanism L7. Hammett equation & its implication to reaction mechanism T2. Numerical problems on Hammett equation	T1: Ch. 4, pg. 97-102. R1: Ch. 39, pg. 1029-1067 Lecture notes	Propose a rationale for the formation of different products under different reaction conditions from same starting materials.
3. Reactive intermediates in organic synthesis	L8-10. Structure, generation, stability and reactions of carbocation, non-classical carbocations L11. Structure, generation, stability and reactions of carbanions L12. Structure, generation, stability and reactions of free radicals, benzyne, carbenes & nitrenes T3 & T4. Exercise problems on different carbon intermediates	T1: Ch. 4, pg. 64-88. R1: Ch 37, pg. 970-999 & Ch 38, pg. 1003-1023	Propose the reaction mechanism involving <i>in situ</i> generation and further reactivity of carbon intermediates
4. Reactivity of aromatic compounds	L13-14. Aromatic electrophilic substitutions and mechanism L15. Aromatic nucleophilic substitutions: addition-elimination L16-17. Aromatic nucleophilic substitutions: elimination-addition T5. Exercise problems on aromatic reactions.	T1: Ch. 5C, pg. 262-284 Ch. 9, pg. 488-503. R1: Ch. 21, pg. 524-541. Ch. 22 pg. 498-528.	Identify the structure of major product(s) in different aromatic reactions and propose a reasoning for their formation of major product under different conditions.
5. Functional group chemistry: Alkyl and aryl halides	L18. Synthesis of alkyl and aryl halides L19-20. Reactions of alkyl and aryl halides and comparison.	T1: Ch. 8, pg. 426-477. Ch. 9, pg. 482-485.	Perform functional group conversion of alkyl/aryl halides to other organic molecules.



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	T6. Exercise problems on application of reactivity of alkyl and aryl halides		
6. Functional group chemistry: Alcohols, phenols and ethers	L21-22. Synthesis of alcohols, diols, phenols L23-24. Reactions of alcohols, diols and phenols L25. Synthesis and reactions of ethers, acid/base catalyzed ring opening of oxirane T7. Exercise problems on application of reactivity of alcohols, phenols & ethers.	T1: Ch. 10, pg. 507-540. Ch. 11, pg. 544-563.	Convert alcohols/ phenols into other organic molecules, and design synthetic schemes for preparing selected commercial molecules such as aspirin, picric acid etc.
7. Functional group chemistry: Carbonyl compounds	L26-27. General methods for the synthesis of aldehydes and ketones L28-29. Reactivity of carbonyl compounds: aldol, crossed aldol reaction, Claisen condensation, Mannich reaction and Wittig reaction. L30-31. Synthesis and reactivity of enolates: Malonate and ethyl acetoacetate L32. Conjugate addition reactions of α,β -unsaturated carbonyl compounds T8-9. Tutorial problems on carbonyl chemistry	T1: Ch. 12, pg. 570-611. R1: Ch 6, pg. 125-140; Ch: 20, pg. 449-468; Ch, 26, pg. 614-652. Lecture notes	Carry out the conversion of carbonyl compounds to different compounds and apply them towards the constructing of C-C, C-N bonds in different molecular environments.
8. Functional group chemistry: Carboxylic acid & derivatives	L33-34. Synthesis & reactions of carboxylic acid and their derivatives L35. A comparative study on reactivity of different acid derivatives T 10. Tutorial problems on carboxylic acid & derivatives.	T1: Ch. 13, pg. 624-648, Ch. 14, pg. 657-685.	Understand the reactivity of acid and their derivatives, and easily inter-convert them.
9. Functional group chemistry: Amines and nitro compounds	L36. Synthesis of amines and nitro compounds L37-38. Basicity and reactions of amines and nitro compounds T 11. Tutorial problems on amines, & nitro compounds.	T1: Ch. 15, pg. 696-736	Convert amines and nitro compounds into various other organic molecules.
10. Carbohydrates	L39. Introduction, classification of carbohydrates, mutarotation. L40. Reactions of hexoses and ketoses (monosaccharides), structure of disaccharides T12. Tutorial problems on carbohydrates.	T1: Ch. 26, pg. 1227-1242, 1245-1253. 1260-1265	Understand a basic knowledge of carbohydrate, and explain the existence of open and closed form in a monosaccharide.



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5. Evaluation Scheme:

Component	Weightage	Duration	Date and time	Comments
Quiz/Assignment	24%	20 min.	TBA	Best 3 of 4, Close book
Attendance Linked Evaluation	10%	20 min.	TBA	Close book
Mid Semester	26%	90 min.	TBA	Close book
Comprehensive	40%	120 min.	TBA	Open book (20%), Close book (20%)

6. Minimum criterion for awarding valid grades: Where a student continued to remain registered in a course but gave the instructor inadequate opportunity to evaluate him/her by absenting himself/herself from quizzes/ tests/ examinations/other components of evaluation, or by appearing in the same for the sake of appearance without applying himself/herself to the task in hand or by submitting a blank script (answer book) this event will be reported as NC (Not Cleared). **Moreover, a student should obtain 30% of the average of the top 3 performers in the class, or 40% of the median marks of the class, whichever is lower to clear the course.** If any student gets the marks lower than the prescribed standard mentioned above, he/she may be awarded NC.

7. Chamber consultation hours: TBA

8. Attendance Policy: As a part of attendance linked evaluation, 2 quizzes (1 before midsem & 1 after midsem) will be held, which will account for 10% weightage. These quizzes will be considered for students with attendance $\geq 50\%$

9. Makeup Policy: Make-up only in genuine emergency/unavoidable cases such as medical admission in hospital or participation in an event to represent the Institute. No make-up to attend marriages or other social functions.

Paritosh Shukla
Instructor-in-charge
CHEM F21